

Holi e-Card in 'C'

NISHANT MITTAL

E-cards are digital greeting cards created on the Web and sent through e-mail. With the festival of colours 'Holi' round the corner, here we present an e-card program for Holi greetings that is written in 'C' language.

Graphics in 'C'

'C' language is a versatile tool in the software world. Graphical user interface (GUI) was a major concern earlier. With 'C' programming language and its graphics utilities, it became the back-end tool for almost all the Micro-soft GUI applications.

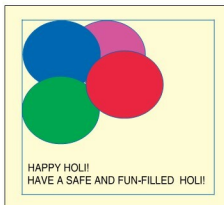
The entire computer monitor screen is made of pixels. Pixels are picture elements—tiny dots of individual colour that make up what you see on your screen. When a graphics program is written in 'C', one actually works with the interface between the monitor and the video graphics array (VGA) card present inside the PC. 'C' has rich built-in graphics functions, allowing programmers to design and develop any graphics picture without much problem.

Software program

The program for the Holi e-card (holi.c) is given at the end of this article.

For the program to work properly, Litt.chr, Euro.chr, Goth.chr, Sans.chr, Trip.chr, Tscr.chr, Egavga.bgi and Simp.chr graphics files are to be stored in the directory where the executable file is located.

Graphics mode initialisation. Normally, the monitor screen seen in DOS is in the text mode. To view graphics, you need to initialise the graphics mode. For this, the graphics drivers need to be initialised on the computer. This is done using the `initgraph` method provided in the `graphics.h` library



Program output

of 'C'.

First, declare two integer-type variables 'gd' and 'gm' for graphics driver and graphics mode, respectively. You can choose any other variable name as desired.

Using `initgraph(&gd,&gm,"c:\\tc\\bgi")` code, provide three arguments to the `initgraph` function—the address of 'gd', the address of 'gm' and the path where graphics or 'bgi' files are located. `Initgraph` initialises the graphics system by loading a graphics driver from the disk (or validating a registered driver) and then putting the system into graphics mode. It also resets all graphics settings (colour, palette, current position, view port, etc) to their default values. `c:\\tc\\bgi` specifies the directory path for the graphics driver (egavga.bgi). If graphics files are not there, `initgraph` looks for them in the current directory. The disk drive and path can be changed as required.

Normally, `initgraph` loads a graphics driver by allocating memory for the driver, then loading the appropriate bgi file from the disk. As an alternative to this dynamic loading scheme, a graphics driver file (or several of them) can be directly linked into the executable program file.

After initialisation of the `initgraph` function, other 'C' functions can be

used to draw shapes such as circle, line and rectangle using the `circle(x-coordinate, y-coordinate, radius)`; syntax. The circle command takes input for an X coordinate (vertical axis) and a Y coordinate (horizontal axis). The last one is the radius of the circle.

Major 'C' functions. The major 'C' functions used in the program are `setcolor`, `moveto`, `outtext` and `outtextxy`.

The `setcolor` function sets the current drawing colour.

The `moveto` function moves the current position to X and Y coordinates.

The `outtext` displays text at the current position. Do not use text-mode functions like 'printf' and 'gotoxy' while working in graphics mode. Also, '\n' or other escape sequence does not work in graphics mode. Ensure that the text doesn't go beyond the screen while using `outtext`.

The `outtextxy` function is for use in graphics mode and displays a string of characters at the specified location.

Testing procedure

To test the program working, proceed as follows:

1. Check whether the declaration and the path given for `initgraph` is defined properly in the holi.c program. The path for "c:\\tc\\bgi" may differ from computer to computer.

2. Compile the holi.c program using Turbo C++ Version 3.0. Generate the holi.exe file using 'Build All' option.

3. Run the holi.exe file. First, the circle animation appears on the screen. If the speed of animation is very fast, change the value in the `delay()` function to decrease it.

4. If the graphics output is stretched, check the maximum and minimum X and Y values using `getmaxx` and `getmaxy` programs, respectively. (The program was tested on a

